

Muhammad Haroon Khan

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EDUCATION

Ph.D. Physics	Aug 2022 – Present	University of Houston, Houston, TX
M.S. Physics	Aug 2022 – Aug 2024	University of Houston, Houston, TX
M. Phil Physics	Feb 2020 – Apr 2022	Quaid-I-Azam University, Islamabad
B.S. Physics	Sep 2015 – Jan 2020	Quaid-I-Azam University, Islamabad

HONORS & AWARDS

❖ Graduate Research Assistantship	2022 – 2027	Higher Education Commission of Pakistan, USA
❖ Research Assistantship	2024 – Present	University of Houston, Houston, TX, USA
❖ TcSUH Graduate Student Travel Support Award	2026 – 2027	Texas Center for Superconductivity, University of Houston, TX, USA
❖ TEX-E Fellowship	2026 – 2027	TEX-E, USA
❖ Best Presentation Award, Experimental Physics Session	2025	University of Houston Physics Research Day, USA
❖ Departmental Fellowship for Academic Excellence	2022	Quaid-i-Azam University, Islamabad

WORK EXPERIENCE

Graduate Student Researcher	Aug 2023 – Present	University of Houston, TX
❖ Supervisor: Dr. Shuo Chen ❖ Investigating high-performance Mg_3Sb_2-based thermoelectric materials for low-grade waste heat recovery and cooling applications through transition-metal doping, defect engineering and phase composition optimization. ❖ Investigating the microstructure, interfaces and crystallographic defects in functional materials, including Hg1223 high-temperature superconductors and transition metal based electrocatalysts using advanced characterization techniques such as TEM, SEM, electron diffraction, EDS and XRD. ❖ Developing novel, low-cost electrocatalysts for green hydrogen production through electrochemical water-splitting, correlating structural properties via XRD and TEM with catalytic performance and stability in industrial-scale water electrolyzer systems.		
Teaching Assistant	Fall 2024	University of Houston, TX
❖ Teaching Assistant in College Physics Lab 1 ❖ Demonstrated to students their weekly lab experiments, conducted weekly office hours, lab report grading, and exam grading.		
Master Student Researcher	Feb 2021 – Apr 2022	Quaid-I-Azam University
❖ Supervisor: Dr. Hassnain Jaffri		

- ❖ Investigated Ba-doped and Ba–Ti co-doped $BiFeO_3$ thin films synthesized via the chemical solution deposition method. Performed Structural and microstructural characterizations using XRD, Raman spectroscopy, SEM and studied optical, electrical and magnetic properties of multifunctional oxide thin films.

RESEARCH PUBLICATIONS

Published

- ❖ **Muhammad Haroon Khan**, Rui Shu, Shuo Chen *et al.*, “Phase Segregation in High Mn-doped Mg_3SbBi Based Thermoelectrics”. *ACS Applied Energy Materials* (2025)
- ❖ Mumtaz, Fiza, **Muhammad Haroon Khan**, and G. Hassnain Jaffari. "Correlation between the composition, phase, band structure, ferroelectric and leakage responses of $Bi_{1-x}Ba_xFe_{1-y}Ti_yO_3$ thin films." *Thin Solid Films* (2022).
- ❖ Publications can be accessed on my [Google Scholar Profile](#).

Manuscripts in Preparation

- ❖ **Muhammad Haroon Khan**, L. Deng, S. Chen *et al.*, “Pressure Induced Edge Dislocations Study of Pristine and Pressure Quenched Hg-1223 Superconducting Materials.” Manuscript in preparation.
- ❖ P. Byaruhanga, **Muhammad Haroon Khan**, Yu Wang, S. Chen *et al.*, “Synergistic Sulfur-Selenium Co-Doping of NiFe LDH on NiSe Nanowires Enables Fast Reconstruction and Enhanced Oxygen Evolution.” Manuscript in preparation

ORAL PRESENTATIONS AND SUMMER SCHOOL

- ❖ **Muhammad Haroon Khan**, Shuo Chen, “ Modulating phase purity boosts thermoelectric performance in high Mn-doped $Mg_3(Sb, Bi)_2$ -based materials ” The 42nd International Conference on Thermoelectrics 2026, Houston TX, USA.
- ❖ **Muhammad Haroon Khan** “ Role of Phase Purity in $Mg_{3-x}Mn_x(Sb, Bi)_2$ Thermoelectric Materials”. 61st TcSUH Student Research Symposium (2026), USA
- ❖ **Muhammad Haroon Khan**, Rui Shu, Shuo Chen “ Phase segregation in high Mn-doped Mg_3SbBi thermoelectric materials”. *American Physical Society & TSAAPT Joint Spring Meeting* (2025).
- ❖ Paul Byaruhanga, **Muhammad Haroon Khan**, Shuo Chen “Investigating Effect of Dual Doping S and Se on NiFe Layered Double Hydroxide Catalyst for Oxygen Evolution Reaction”. *Electrochemical Society Meeting* (2025).
- ❖ 4th Summer School on Machine Learning For Electron Microscopy June 22-26, 2026 **University of Tennessee, Knoxville, USA**
- ❖ Tech Bootcamp on Robophysics Summer 2024 **Georgia Institute of Technology, GA, USA**
- ❖ IPOLS Scientific Programming in Python Bootcamp Summer 2024 **Emory University, GA, USA**
- ❖ International School on Rietveld Refinement & Paid Distribution ([ISBARM-22](#)) Summer 2022 **National Centre for Physics (NCP), PAK**

RELAVENT SKILLS

Material Synthesis and Processing

- ❖ Bulk thermoelectric materials synthesis, thermoelectric module assembly, catalyst synthesis for alkaline water electrolysis, thin-film synthesis via sol-gel deposition, high-temperature annealing and thermal processing.

Electron Microscopy

- ❖ Advanced electron microscopy techniques including, TEM, STEM, HRTEM, SAED, STEM-EDS, SEM, FIB, PIPS II ion polishing, dimpling, mechanical polishing and cross-sectional TEM specimen preparation.

Material Characterizations

- ❖ Structural and functional characterizations using X-Ray Diffraction, Rietveld refinement SEM, TEM, Raman Spectroscopy, UV-Visible Spectroscopy, XPS, magnetic measurements, Hall measurements, thermal and electrical measurements, thermoelectric device measurements, and dielectric measurements.

Programming and Software

- ❖ ImageJ, FullProf / Rietveld Refinement, Origin Pro, VESTA, Strain++, Python, MATLAB, C++, HTML, Machine learning for electron microscopy

LEADERSHIP AND VOLUNTEERING EXPERIENCE

❖ TEX-E Fellow	2026 – 2027	TEX-E
❖ Vice Chair of Mentorship Committee	Fall 2025 – Present	UH Energy Coalition
❖ Assistant Session Chair	May 17–21, 2026	Internation Thermoelectric Conference 2026, Houston
❖ NSM Grad. Student Council Representative	2024 – 2025	Department of Physics at the University of Houston
❖ Vice Chair of Sustainability Committee	Spring 2025	UH Energy Coalition
❖ Judge in Physics and Astronomy category at Science & Engineering Fair Houston	2022 – 2026	SEFH, Houston